Non-negative Supermartingale Theorem

Short Notes by Iman Taghaddosinejad

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Non-negative supermartingales arise naturally in dynamic macroeconomic models through Euler inequalities and provide a useful way to establish the long-run convergence of endogenous stochastic processes.

Theorem 1 (Non-negative supermartingale convergence theorem)**.** *Let X_t be a non-negative supermartingale such that $X_t \geq \mathbb{E}_t[X_{t+1}]$ and $X_t \geq 0$ for all $t \geq 0$. Then, there exists a finite random variable $X_\infty \geq 0$ such that $X_t \to X_\infty$ a.s. as $t \to \infty$. Moreover, we have $\mathbb{E}[X_\infty] \leq \mathbb{E}[X_0]$.*

Remark 1 (Intuition)**.** *Theorem 1 states that if a random process is a non-negative supermartingale then it converges almost surely (along any sequence with positive probability) to a finite non-negative (random variable) limit. The process will not diverge to $\pm\infty$ or oscillate over some positive range $b - a$ with $b > a > 0$ infinitely often.*

The proof of Theorem 1 intuitively works in three steps. First, we rule out divergence to $\pm\infty$ using basic properties of the process itself. Second, we rule out persistent oscillations by showing that the process cannot upcross any interval of positive width infinitely often. The key observation is that if $\liminf_{t\to\infty} X_t < \limsup_{t\to\infty} X_t$, then there must exist some interval $[a, b]$ of positive width lying strictly between them that the process upcrosses infinitely often. Thus, ruling out infinitely many upcrossings rules out any positive gap between the lim inf and lim sup. Finally, having ruled out both divergence and persistent oscillation, we obtain $\liminf_{t\to\infty} X_t = \limsup_{t\to\infty} X_t < \infty$ a.s. and hence X_t converges to a finite limit almost surely.

Proof *(Sketch proof of Theorem 1).* Let X_t be a non-negative supermartingale. First we establish the process cannot diverge to $\pm\infty$. Since $X_t \geq 0$

for all $t \geq 0$, we immediately have $\liminf_{t\to\infty} X_t \geq 0$ a.s. Moreover, since $X_t \geq \mathbb{E}_t[X_{t+1}]$, taking the unconditional expectation of both sides and using the law of iterated expectations (LIE) yields $\mathbb{E}[X_t] \geq \mathbb{E}[X_{t+1}]$. Therefore, $\infty > \mathbb{E}[X_0] \geq \mathbb{E}[X_1] \geq \mathbb{E}[X_2] \geq \cdots \geq 0$. Hence, $\mathbb{E}[X_t]$ is a decreasing sequence bounded below by zero and therefore converges to some finite limit. Moreover, the sequence is uniformly bounded above, $\sup_{t\geq 0} \mathbb{E}[X_t] = \mathbb{E}[X_0] < \infty$. Together with non-negativity, this rules out the event $\mathcal{A} := \{\omega : X_t(\omega) \to +\infty\}$, having positive probability. Otherwise, paths in $\mathcal{A}$ would make an arbitrarily large positive contribution to the ensemble mean, while $X_t \geq 0$ prevents this contribution from being offset by negative realizations, contradicting the finite uniform bound on $\mathbb{E}[X_t]$. Hence, $\mathbb{P}(X_t \to +\infty) = 0$.

Next we rule out persistent oscillations over any interval of positive width. Let $a < b$ and suppose that $0 \leq \liminf_{t\to\infty} X_t < a < b < \limsup_{t\to\infty} X_t$. Let U_T denote the number of completed upcrossings of the interval $[a, b]$ by time T where $0 \leq U_1 \leq U_2 \leq \cdots \leq U_T$. Further define the auxiliary process $H_t \in \{0, 1\}$ such that $H_t = 1$ if X_t is in the midst of an upcrossing of $[a, b]$ and $H_t = 0$ otherwise. At any time $T \geq 0$, the process has therefore completed U_T upcrossings and may additionally be in the midst of at most one unfinished upcrossing. Let s_i and τ_i denote the start and completion times of the i-th completed upcrossing, respectively, so that $X_{s_i} \leq a$, $X_{\tau_i} \geq b$, and hence $X_{\tau_i} - X_{s_i} \geq b - a$. If an unfinished upcrossing remains at time T, let s denote its starting time, so that $X_s \leq a$. Then we can define the sum of all contributions of upcrossings

$$
\begin{aligned}
G_T &= \sum_{t=0}^{T-1} H_t(X_{t+1} - X_t) \\
&= \underbrace{H_0(X_1 - X_0) + \cdots + H_{s-1}(X_s - X_{s-1})}_{\text{completed upcrossings}} + \underbrace{H_s(X_{s+1} - X_s) + \cdots + H_{T-1}(X_T - X_{T-1})}_{\text{unfinished crossing}} \\
&= \underbrace{\sum_{i=1}^{U_T} (X_{\tau_i} - X_{s_i})}_{\geq (b-a)U_T \geq 0} + \underbrace{X_T - X_s}_{\geq -a} \\
&\geq (b - a)U_T - a,
\end{aligned}
$$

where the last inequality follows because $X_T \geq 0$ and $X_s \leq a$. Furthermore,

we can derive an upper bound on the ensamble mean of G_T:

$$\begin{aligned}
\mathbb{E}[G_T] &= \sum_{t=0}^{T-1} \mathbb{E}\big[H_t(X_{t+1} - X_t)\big] \\
&= \sum_{t=0}^{T-1} \mathbb{E}\Big[\mathbb{E}_t\big(H_t(X_{t+1} - X_t)\big)\Big] \\
&= \sum_{t=0}^{T-1} \mathbb{E}\Big[\underbrace{H_t}_{\geq 0}\underbrace{\mathbb{E}_t(X_{t+1} - X_t)}_{\leq 0}\Big] \\
&\leq 0,
\end{aligned}$$

where the second equality uses the LIE, the third equality uses the fact that H_t is known at time t, and the last inequality uses the supermartingale property. Combining the pathwise inequality for G_T with the ensemble mean inequality yields $\mathbb{E}[U_T] \leq \frac{a}{b-a} < \infty$ for all $T \geq 0$. Therefore, $\sup_{t\geq 0} \mathbb{E}[U_t] \leq \frac{a}{b-a} < \infty$. Since $U_T \geq 0$, this uniform bound rules out any positive-probability set of paths along which the number of completed upcrossings grows without bound. Thus, for the event $\mathcal{B} := \{\omega : U_T(\omega) \to \infty\}$, we have $\mathbb{P}(\mathcal{B}) = 0$. Hence, with probability one, X_t can complete only finitely many upcrossings of any fixed interval $[a, b]$ of positive width. This contradicts the assumption that $\liminf_{t\to\infty} X_t < \limsup_{t\to\infty} X_t$ since any positive gap contains an interval of positive width that must be upcrossed infinitely often. Therefore, $0 \leq \liminf_{t\to\infty} X_t = \limsup_{t\to\infty} X_t < +\infty$ a.s., and hence $X_t \to X_\infty < \infty$ a.s. as $t \to \infty$. □

Remark 2 (Null sets, countability, and rational intervals)**.** *A subtle point in the upcrossing argument is that we do not establish that the set of paths which upcross a fixed interval $[a, b]$ infinitely often is empty. Rather, for each fixed $a < b$, we establish the weaker result that this set has probability zero. This distinction matters, as we do not rule out the existence of probability-zero paths.*

To pass from the result for each fixed interval to almost-sure convergence, let $\mathcal{C} := \{\omega : \liminf_{t\to\infty} X_t(\omega) < \limsup_{t\to\infty} X_t(\omega)\}$, and define the event $\mathcal{B}_{a,b} := \{\omega : U_T(\omega) \to \infty,$ with U_T on $[a, b],\ a, b \in \mathbb{Q}\}$. For every path $\omega \in \mathcal{C}$, $\exists$ rationals $a < b$ satisfying $\liminf_{t\to\infty} X_t < a < b < \limsup_{t\to\infty} X_t$,

and hence $\omega \in \mathcal{B}_{a,b}$*, Therefore,*

$$\mathcal{C} \subseteq \bigcup_{\substack{a,b\in\mathbb{Q},\\ a<b}} \mathcal{B}_{a,b}.$$

Moreover, $\mathbb{Q}$ *is countably infinite, such that the collection of rational pairs* (a, b) *is also countable. Therefore,*

$$\mathbb{P}(\mathcal{C}) \leq \mathbb{P}\left(\bigcup_{\substack{a,b\in\mathbb{Q},\\ a<b}} \mathcal{B}_{a,b}\right) \leq \sum_{\substack{a,b\in\mathbb{Q},\\ a<b}} \mathbb{P}(\mathcal{B}_{a,b}) = 0,$$

where the first inequality uses monotonicity of probability, while the second uses countable subadditivity since the union is countable. Thus, exceptional paths may exist, but the set containing all of them has probability zero. The use of $\mathbb{Q}$ *ensures that every positive gap formed by a path in* $\mathcal{C}$ *contains a rational interval, while being countable enough that the union of the corresponding probability-zero events remains a probability-zero event.*